LLM Hallucination Mitigation: Leveraging Mechanistic Signals in Reinforcement Learning Pipelines

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Abstract

We investigate hallucination mitigation in LLMs by integrating an intermediate mature layer consistency signal, inspired by DoLa, which showed that agreement between early and late transformer layers is tied to more stable and factual predictions. We incorporate this signal into reinforcement learning by applying DoLa style layer contrast during PPO and GRPO rollouts and evaluate its effect on GSM8K using a Qwen2.5-3B backbone. We train four models: PPO, PPO + DoLa, GRPO, and GRPO + DoLa. Preliminary results shows a small improvement under the flexible-extract exact-match metric for PPO + DoLa. These findings suggest that mechanistic layer-consistency signals may help guide RL training toward more reliable reasoning.

1 Introduction

The alignment of LLMs with objectives such as factuality and reliability remains a central challenge in their development. A significant impediment to their deployment in high-stakes applications is the phenomenon of hallucination, wherein models generate fluent but factually unsubstantiated content. This behavior is an intrinsic artifact of standard training paradigms that optimize for statistical plausibility over factuality.

Prevailing mitigation strategies can be broadly categorized into training-time and inference-time interventions. During training, policy optimization algorithms such as Group Relative Policy Optimization (GRPO) [10] are employed to align model outputs with a reward signal, which can implicitly encourage factuality. However, it has been observed that optimizing for reasoning or helpfulness can paradoxically increase a model's propensity to hallucinate, this phenomenon is termed as the Hallucination Dilemma [6]. At inference, methods such as Retrieval Augmented Generation (RAG)[5] ground responses in external knowledge, though their efficacy is contingent on retrieval quality.

A distinct line of research has focused on the mechanistic properties of transformer architectures, revealing that different layers encode increasingly refined representations of the next token distribution. DoLa [1] leverages this observation by contrasting the logits of "premature" intermediate layers with those of the final "mature" layer to identify and suppress unstable or hallucination prone predictions at inference time. While effective, DoLa operates purely as a post hoc decoding method and does not modify the model's parameters. This raises the question of whether the layer contrast signal can be embedded directly into the RL framework to guide the model toward more reliable behavior.

In this work, we investigate whether DoLa's layer contrast signal can influence reinforcement learning when incorporated directly into the rollout process. Rather than using DoLa solely at inference, we integrate its intermediate–final layer contrast into the PPO and GRPO training pipelines and examine how this affects the learned policy. By training models with and without this signal on GSM8K [2], we assess whether exposing the policy to layer level disagreement during rollouts can guide learning toward more stable and reliable behavior.

2 Motivation

The reliable deployment of LLMs is limited by their tendency to generate non factual content, commonly referred to as hallucination. Existing mitigation strategies are often reactive or rely on external supervision, making them difficult to scale. This work is motivated by the idea that internal model signals specifically, the contrast between intermediate and final transformer layers may provide useful guidance during training. We therefore investigate whether incorporating a DoLa inspired layer contrast signal into the reinforcement learning process can influence learning dynamics and encourage more reliable behavior. By embedding this mechanistic signal within PPO and GRPO training on GSM8K, we examine its effect on model stability and hallucination tendencies.

3 Related work

Research in hallucination mitigation for Large Language Models can be broadly categorized based on the point of intervention in the model lifecycle: inference time guidance, external knowledge grounding, and training time policy optimization. Inference time approaches aim to improve output factuality without modifying model parameters. These include structured prompting techniques like Chain of Thought (CoT) and post hoc verification methods such as Chain of Verification[3]. A parallel direction focuses on the mechanistic underpinnings of the transformer architecture. The DoLa[1] decoding strategy, for instance, contrasts logit distributions between early and late layers at inference time to suppress non factual tokens. While these methods are effective, they are fundamentally reactive and do not alter the model's intrinsic generative distribution.

Training time interventions seek to align model parameters with factuality objectives, primarily through reinforcement learning (RL). Group Relative Policy Optimization (GRPO)[10], a memory efficient policy gradient method, has shown considerable success in enhancing LLM reasoning capabilities. The initial formulation of GRPO, however, relies on a sparse, outcome based reward, which provides a weak learning signal for complex, multi step tasks, a limitation known as abrupt reward vanishing[4]. This limitation has motivated a clear progression toward developing more nuanced forms of process supervision. An initial class of solutions introduces external signals to enrich the learning process. Methods such as MGRPO[4] create multistage, self correction loops, while factuality aware approaches like FSPO[6] and KnowRL[9] incorporate explicit fact checking against an external knowledge source. A subsequent line of inquiry has focused on leveraging internal model states for supervision. Early work in this direction, such as CoDaPO[11], utilizes the model's output confidence to reweight the advantage signal. More recently, the RISE framework[7] operationalizes self verification by jointly training an explicit verifier head within the RL process to provide an online reward signal.

Our work contributes a novel internal signal derived from model mechanics and leverages a state of the art credit assignment mechanism. In contrast to factuality aware RL methods that depend on external verifiers, our approach is self supervised. Distinct from self verification frameworks such as RISE, which employ an explicitly trained verifier head, our process level signal is derived implicitly from the model's internal mechanics. Furthermore, unlike DoLa, which operates as a post hoc decoding strategy, our method constitutes a direct training time intervention. The proposed framework thereby merges mechanistic interpretability with recent advances in policy optimization to train an intrinsically more factual generative model.

4 Methodology

Our objective is to study whether DoLa's layer contrast signal can influence reinforcement learning when incorporated directly into the rollout generation process. Instead of using DoLa solely as an inference time decoding strategy, we enable its intermediate–mature layer contrast during the PPO and GRPO rollouts that define the training trajectories. This allows us to examine how exposing the policy to DoLa modified token distributions affects optimization on GSM8K.

4.1 Problem Setting

We consider reinforcement learning for mathematical reasoning, where a policy π_θ generates a sequence $y = (y_1, \ldots, y_T)$ in response to a problem x. In the standard GSM8K setup, the reward is sparse and depends only on whether the final numeric answer is correct, providing limited guidance for multi-step reasoning. In this work, we do not modify the reward itself. Instead, we alter the distribution used to generate trajectories by applying DoLa's contrast between intermediate and final transformer layers during sampling. This integration allows us to investigate whether layer-level disagreement, surfaced during rollouts, can influence the behavior and stability of PPO and GRPO during training.

4.2 Mechanistic Layer–Agreement Signal

For each layer ℓ and token position t, let $h_t^{(\ell)} \in R^d$ denote the last-token hidden state and let $W \in R^{V \times d}$ be the shared output projection. We form layer-specific logits and distributions

$$L_t^{(\ell)} = W h_t^{(\ell)}, \qquad P_t^{(\ell)} = \mathrm{softmax}(L_t^{(\ell)}).$$

We treat the final decoder block as the *mature* layer ℓ_m and define a set $\mathcal{S}$ of candidate *premature* layers.

Layer disagreement. For each premature layer $\ell \in \mathcal{S}$, we quantify mismatch with the mature distribution using Jensen–Shannon divergence:

$$A_t(\ell) = \mathrm{JS}\big(P_t^{(\ell_m)} \,\|\, P_t^{(\ell)}\big), \qquad \mathrm{JS}(P \,\|\, Q) = \tfrac{1}{2}\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\mathrm{KL}(Q \,\|\, M), \quad M = \tfrac{1}{2}(P + Q).$$

Static vs. dynamic pairing. A static configuration fixes a premature layer ℓ_p in advance. The dynamic variant selects, at each token position,

$$\ell_p(t) = \arg\max_{\ell \in \mathcal{S}} A_t(\ell),$$

i.e., the layer whose distribution most disagrees with the mature layer.

4.3 DoLa Contrast During Rollout Generation

To incorporate layer contrast into training trajectories, we apply a DoLa-style logits processor at every decoding step inside the HF `generate` call. During generation, forward-hooks on each transformer block cache the hidden state $h_t^{(\ell)}$ for all layers in the candidate set $\mathcal{S}$. Given the mature logits $L_t^{(\ell_m)}$ provided by the model, we compute a premature distribution $P_t^{(\ell)}$ using the LM head and select $\ell_p(t)$ via the dynamic JS rule described above.

A relative-top mask is applied to the mature logits to restrict sampling to the high-probability support, after which DoLa returns modified scores $S_t = \tilde{L}_t^{(\ell_m)} - \log P_t^{(\ell_p(t))}$. These scores replace the model's own logits during sampling, producing layer-adjusted rollouts used by PPO and GRPO.

5 Experimental Setup

We assess the effect of DoLa-modified rollouts on Proximal Policy Optimization (PPO) and Grouped Regularized Policy Optimization (GRPO) using `Qwen2.5-3B-Instruct` [8] as actor and critic. GSM8K prompts follow the model's chat template and enforce the " `<number>`" answer format to avoid zero-reward failures. Rollouts are generated via HuggingFace `generate()` in VERL, where DoLa functions as a logits processor in either (i) a static form: using a fixed premature layer or (ii) a dynamic form: that selects the premature layer with maximal disagreement from the mature logits; an optional relative-top mask prunes low-probability mature tokens. RL training uses the standard GSM8K correctness reward without auxiliary shaping. Performance is measured using the flexible-extract and strict-extract exact-match metrics.

GSM8K (Exact Match)	PPO	PPO + DoLa	GRPO	GRPO + DoLa
Flexible-extract	0.0000	**0.0015** $\pm$ 0.0011	0.0000	0.0000
Strict-match	0.0000	0.0000	0.0000	0.0000

Table 1: Preliminary GSM8K exact-match accuracy for PPO and GRPO, with and without DoLa-modified rollouts. **Take-away:** Introducing DoLa during PPO rollouts yields a small but measurable improvement under the flexible-extract metric, while all strict-match scores remain near zero. The uniformly low baselines reflect the difficulty of GSM8K in our constrained training setup (short sequences, small group size, strict answer formatting) (see Appendix .1), which likely masks larger underlying differences between methods.

6 Preliminary Results

Analysis of Initial Results As shown in Table 1, applying DoLa during PPO rollouts yields a small improvement under the flexible-extract metric, suggesting that layer-contrast signals can influence the training trajectory. However, exact-match accuracy remains near zero across all models. This is likely due to constraints of our setup: short rollouts, limited GPU budget, and GSM8K's strict answer-format sensitivity which depress baseline performance and reduce the observable effect size (see Appendix .1). These findings should therefore be viewed as *directional* rather than *absolute* gains. Under the current constraints, the experiments cannot fully reflect the impact of integrating layer-contrast signals into PPO/GRPO; a more complete evaluation will require alleviating these constraints within the ongoing iteration of the project.

7 Plan of Work

Refine prompt structure and rollout framework. Many trajectories received zero reward due to malformed final answers (e.g., missing or incorrect " `<number>`" markers) or truncated generations. We will revise the prompting template to enforce stable answer formatting and reduce such reward dropouts. In parallel, we will explore alternative RL frameworks that offer access to token-level logit than the current HF+VeRL integration.

Leverage DoLa's early-exit signal to enhance exploration. If prompt refinements do not improve reward density, we will investigate how DoLa's layer-disagreement spikes an indicator of model uncertainty can be used to widen the sampling distribution during GRPO rollouts. The goal is to exploit this signal to promote richer exploration rather than allowing GRPO to collapse into low-entropy trajectories.

8 Conclusions

We explored whether mechanistic layer-agreement signals capturing the contrast between intermediate and mature transformer layers can be integrated directly into reinforcement learning. By embedding a DoLa-style logits modification into PPO and GRPO rollouts, we show that interpretability derived signals can interact with policy optimization. Preliminary GSM8K results reveal a small directional gain for PPO+DoLa, indicating that exposing the policy to layer-level disagreement can subtly shape learning dynamics.

References

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.1 Limitations

Our study faces several practical and methodological limitations. First, integrating DoLa within HuggingFace generation increases memory pressure, restricting GRPO group size and maximum sequence length. Second, effective training requires at least four GPUs, and our access to compute constrained the number of configurations and ablations we could explore. Third, GSM8K rewards are highly sensitive to answer formatting; even minor deviations from the required "" pattern lead to zero reward. This sensitivity likely contributes to the lower-than-expected baseline scores in our runs, which in turn suppresses the apparent gains of the proposed DoLa-integrated variants. Finally, the current VERL framework has limited support for HuggingFace-style rollouts— ven though HF is the only interface that exposes per-token logits needed for DoLa suggesting that a different RL framework may be more suitable for future work.